Rajat Shashidhar Hegde

co24btech11020@iith.ac.in | rajathegde2006@gmail.com | +91 8088952606 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.78
Shantiniketan Science PU College	Class XII (Karnataka State Board)	2024	95.50%
SKB English Medium High School Kansur	Class X (Karnataka State Board)	2022	99.84%

SCHOLASTIC ACHIEVEMENTS

Ranked among the Top 1500 in Google Big Code 2026

Qualified Stage 2 of AlgoUniversity Technology Fellowship 2025, ranking in the top 8% among 2,50,000+ participants

Codeforces: 1568 (Specialist) | LeetCode: 1938 (top 3.6%) | CodeChef: 1624 (3 star)

WORK EXPERIENCE

Web Development Intern - Eventzx

- Developed and deployed two full-stack web applications using React, Supabase, and Vercel with a shared architecture.
- Built authentication, friend management, one-to-one & group messaging, communities, and opportunity posting.
- Used **Supabase Auth** and **PostgreSQL** to implement secure authentication and low-latency messaging.
- Improved code reusability by designing shared frontend components and backend modules across both applications.
- Collaborated in a startup environment to rapidly develop, test, deploy, and iterate on **production-ready** features.

PROJECTS

AI Resume Ranking System -

Tech: Python, BGE, BM25, Cross-Encoder, Semantic Search

- Built a resume ranking pipeline combining **BGE embeddings**, BM25, structured scoring, and Cross-Encoder reranking.
- Designed a multi-stage retrieval pipeline that reranks top 100 candidates from stage 1 while meeting runtime constraints.
- Optimized ranking of **100K pre-indexed resumes** in under 5 minutes on CPU using precomputed embeddings.
- Improved ranking accuracy by fusing semantic similarity, lexical retrieval, and structured scoring along multiple dimensions.

Natural Language Video Investigation System -

***Tech**: Python, YOLO, ByteTrack, CLIP, FAISS, VideoMAE*

- Built a semantic video retrieval pipeline that locates relevant video timestamps from natural language queries.
- Integrated **YOLO** and **ByteTrack** for object detection and multi-object tracking across video frames.
- Combined **CLIP**, **FAISS**, and **VideoMAE** to perform semantic retrieval using visual embeddings and action recognition.
- Enabled efficient search over long-form videos through a multi-stage indexing and vector search pipeline.

OS-Level Performance & Energy Management | Github link

***Tech**: C++, Linux, Intel RAPL, Reinforcement Learning*

- Built a **Linux framework** to monitor CPU utilization, IPC, temperature, I/O wait and power metrics for dynamic

optimization.
- Developed an **RL agent** that modifies kernel policies across multiple workload types based on runtime behavior.
- Reduced **power consumption by ~10%** while maintaining latency within **~2%** during workload execution.
- Created a modular monitoring pipeline with a real-time visualization dashboard for metrics and workload analysis.

Layer 4 TCP Load Balancer | Github Link

***Tech**: C++, TCP, epoll, Docker*

- Built an event-driven **Layer 4 TCP load balancer** using non-blocking sockets and **epoll**.
- Implemented **Round Robin, Least Connections, and Weighted Round Robin** scheduling algorithms.
- Added automatic backend **health checks** to detect failures and allow backend recovery without interrupting the load balancer.
- Developed connection management, partial I/O buffering, structured logging, metrics endpoint, and Docker deployment.

Intellecta - Developer Social Platform | Github Link

***Tech**: MERN, Socket.IO, Supabase*

- Built a developer-focused social platform with authentication, posts, comments, likes, friend requests, and user search.
- Implemented **one-to-one and group messaging** using **Socket.IO** with syntax-highlighted and formatted code sharing.
- Developed secure backend APIs supporting **user search**, real-time interactions, friend management and social feeds.
- Built a responsive **MERN** application with secure user authentication real time data synchronization.

SKILLS

Programming languages: C, C++, JavaScript, Python, Go, SQL, HTML, CSS

AI/ML: PyTorch, TensorFlow, OpenCV, FAISS, YOLO, ByteTrack, Semantic Search, NumPy, Pandas

Information Retrieval: Semantic Search, Vector Search, BM25, Cross-Encoder Reranking, BGE Embeddings

Systems & Backend: Linux, Docker, Git, REST APIs, MongoDB, PostgreSQL

RELEVANT COURSES

Maths: Introduction to Probability, Elementry Linear Algebra, Transform Techniques, Calculus I & II

Computer Science: Data Structure and Applications, Intro to Programming, Operating System I & II, Digital Circuits, Intro to HDL

POSITIONS OF RESPONSIBILITY

core member, Lambda - software development club (june 2025 - may 2026)

web domain coordinator, FCC - Fincance and Consulting club (june 2025 - may 2026)

EXTRACURRICULAR

Competed in multiple state-level chess tournaments.

Participated in regional Badminton tournaments